

Selin Karaca

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EDUCATION

Columbia University, Fu Foundation School of Engineering and Applied Science Expected: May 2028 | New York, NY
BS in Computer Science, Minors in Applied Math and Entrepreneurship & Innovation, GPA: 4.09, Named Scholar, Full Grant Scholarship

Related Coursework: Natural Language Processing, Analysis of Algorithms, Data Structures and Algorithms, Advanced Programming, Fundamentals of Computer Systems, Discrete Mathematics, Linear Algebra, Probability for Engineers

WORK EXPERIENCE

Savanah.ai, Startup New York, NY
AI Personalization Intern Jun 2026 – Present

- Designed the company's first personalization system, mapping customer behavior and profile signals to dynamically tailored storefront visuals.
- Specified an unsupervised segmentation pipeline (behavioral embeddings, UMAP, HDBSCAN, k-means) to replace hand-written rules with data-derived cohorts.
- Defined pilot data requirements and the A/B test design used to measure CTR, add-to-cart rate, and conversion lift.

Citadel Securities New York, NY
Women's Ignite Trading Week Jan 2026

- Selected for Citadel Securities' competitive trading program; traded simulated markets under time pressure and explicit risk limits.
- Translated market signals into trade theses with quantified risk-reward, sizing rationale, and pre-defined downside controls.

Columbia University, Discrete Mathematics; Fundamentals of Computer Systems New York, NY
Teaching Assistant Jan 2026 – Present

- Teach C programming, memory management, and computer architecture to 300+ students across two core CS courses, alongside proof techniques and induction.
- Lead office hours and discussion sections; grade assignments and exams across theory and systems tracks.

Cool Digital; Efilli, Startup İzmir, Turkey
Software & Product Management Intern Jun – Jul 2025

- Shipped cookie-consent banner interfaces to production for live client websites, built in JavaScript, Vue.js, and React.
- Debugged frontend flows end to end — API requests, auth tokens, login state, routing — resolving user-facing bugs before release.
- Contributed to an AI subproject and prototyped product concepts in Figma, working daily in Git and Chrome DevTools.

RESEARCH

Data Analytics and Processing Lab (DAPLab), Columbia University New York, NY
Undergraduate Research Assistant Jun 2026 – Present

- Build Agent Olympiad, a benchmark for evaluating multi-agent LLM reasoning on Olympiad-level problems; work targeted for submission to ICLR.
- Engineer Python pipelines converting official problems, solutions, and rubrics into structured, machine-gradable benchmark data.

Practice and Research in AI for Science and Education (PRAISE) Lab, Columbia University New York, NY
Undergraduate Research Assistant May 2026 – Present

- Contribute to Automatic Symbolic Law Discovery, applying PyTorch deep learning and computer vision models to recover scientific laws from experimental data.

Complex Resilient Intelligent Systems (CRIS) Lab, Columbia University New York, NY
Undergraduate Research Assistant Jan 2026 – Present

- Investigate semantic emergence in language-model embeddings across model scale, architecture, and training regime; findings support a manuscript in preparation.
- Built label-free PyTorch analysis pipelines (cosine similarity, BERTopic, nearest-neighbor retrieval, anisotropy) over 500 books and 22 model checkpoints, running training and inference on GPU cluster resources.

PROJECTS

Todi, AI-powered learning platform for students with learning differences

Cofounder & Developer

- Built and launched a React and Node/Express learning platform now used by 2,000+ students with learning differences.
- Own the full stack end to end — accessibility-first React UI, Express API, data model, and deployment.

LEADERSHIP & ACTIVITIES

Head of Corporate Relations, Turkish Students Association Global

Lead corporate partnership and sponsorship outreach for a global student network, building company relationships for events, speakers, and career programming.

Y Combinator Startup School, 2026

Completed Y Combinator's founder program on customer discovery, product-market fit, and early-stage growth, applying the framework to Todi's product roadmap and user acquisition strategy.

Global Career Accelerator, Coding for Data Track

Selected by Columbia Engineering; analyzed Intel sustainability KPIs in SQL for a client analytics project.

Microsoft Girls in AI Hackathon

Built a sustainable-agriculture solution using Azure object recognition and presented on responsible AI deployment.

Clubs: Application Development Initiative, Girls Who Code, CAIAC Technical AI Safety Fellowship, Women in CS

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, MATLAB, R

ML & Systems: PyTorch, NumPy, scikit-learn, GPU/cluster training, Linux, Git, deep learning, NLP, computer vision, embeddings, BERTopic, UMAP, HDBSCAN

Web & Tools: React, Node.js, Express, Vue.js, REST APIs, Figma, Chrome DevTools, Microsoft Azure